Augmented Reality Solution for Retail using Visual Commerce Engine

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***Abstract*—Currently, the e-commerce platforms or websites are designed such that users search for products and they are not "store centric" rather product centric. Browsing products over such websites does not add any value to the stores. The below abstract shows the method and system of a Visual Commerce Engine. In this system, stores can come online and showcase their products and improve their branding in the marketplace. This system works best for stores where visual appeal plays a vital role. We are bridging the gap of a faraway shopper with the store, where the shopper can visit the store online, view the products (as if in store) and buy them on the spot in the high quality 360° virtual tour. We have added many shopping features (built using our software toolsets) which will give the shopper an interactive real experience in the store**

Keywords- Visual Commerce, Augmented reality for retail, Customer Engagement, Virtual guided tour for retail shopping

I. INTRODUCTION

Our Augmented Reality Commerce technology blends the trust of a physical store with the convenience of an online store. This helps the stores to go to their customers rather than customers coming to them. Our technology not only enables online commerce, it also drives footfalls to offline stores. We have developed an augmented reality technology where you can embed information in form of text, image & video over virtual tour and provide this solution to the store users so that they experience an unique way of shopping online.

Fig 1: Visual Commerce Platform

One of the key areas where this Visual Commerce solution can be implemented is in the field of Arts & Crafts. AR Solution for retail is an effective mechanism to promote local arts, crafts and artisans. The AR solution enables to set a platform wherein all local niche businesses in the field of arts, crafts and other unique products can subscribe, promote them and sell online. The solution helps to digitize their products and take them to the online world exposing them to the unlimited opportunities.

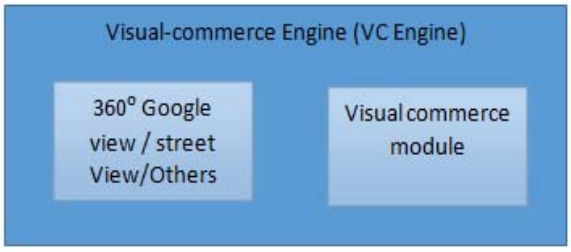

Fig 2: Visual Commerce Engine (VCE)

II. TECHNOLOGY COMPONENTS

Visual-commerce engine consists of mainly two main modules:

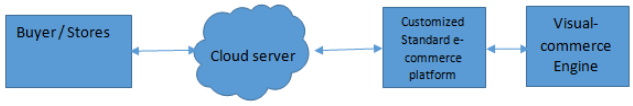

Fig 3: Cloud server for VCE

1) Visual commerce Engine :

Visual commerce Engine is currently built upon Google business view/street view (compatible with any other 360° view) with our proprietary features which enables shopping experience in the high quality 360° virtual tour. The features toolsets added are as follows:

a) Navigation (Direction and Sectioning)
b) Enquiry/Share tool
c) Product details
d) Hotspot technology

Using these feature tools, we can mark/tag the products on the Google business photo /street view iFrame and place them on sale/buy. Those products not on sale can be enquired with the store back end for availability and other queries that customer wish can ask. This simulates the customer engagement with the stores in a real scenario where customer visits physical store and queries for information on products. This engine also enables customers to share the product image on social media. In this way, store promotion and advertising is done. We provide a

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marketplace or platform to multiple stores so that they do not get lost in today's e-commerce wars.

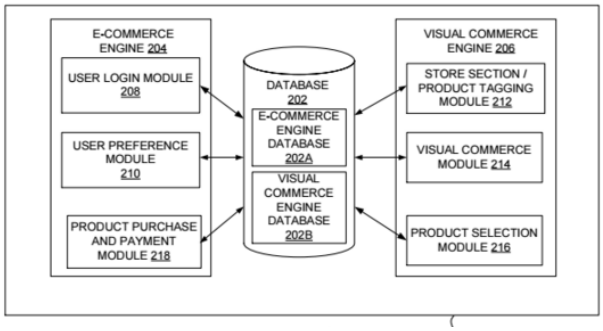

Fig 4: Block diagram of VCE

The above figure shows the modules involved within the visual commerce engine and its interaction with the standard ecommerce engine. There are different modules involved in visual commerce engine such as

a) Store section/product tagging module: used to create various sections in a virtual tour. The products seen in the tour are tagged using algorithm developed specifically to work for virtual tours.
b) Visual commerce module: This contains the features added on the virtual tour to make it interactive for the users such as enquiry, share, auto rotate etc.
c) Product selection module: Contains the business logic to map the hotspots to the products in the ecommerce system.

There is common database used for both the engines to exchange data across so that updation happens real time.

The ecommerce engine contains all the standard modules required for a ecommerce system. We will not go in detail in it.

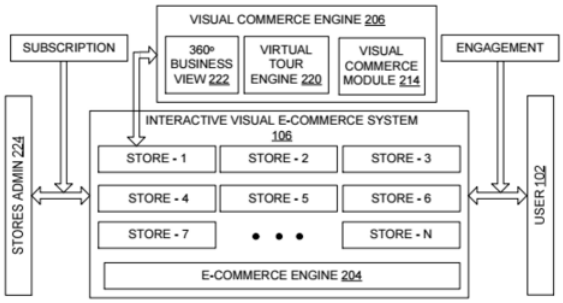

Fig 5: Data flow across platform

The above figure shows data flow across the engine modules in the visual commerce platform. The stores are created in the visual commerce system and interacts with visual commerce engine to provide the visual commerce shopping experience. To control the engine, different modules like store admin etc will be used.

2) Customized Standard e-commerce platform:

Customized Standard e-commerce platform is basically built upon available framework which uses Themes for UI, Modules for various functionalities and Integration with various third party compatible modules.

a) Themes: We have developed our own customized UI for our visual–commerce website. The theme is customized by adding or editing the PHP, HTML and CSS.

b) Modules: Various modules have been incorporated to provide the users seamless way to shop. Modules such as order management, product details, product listing, and delivery management have been customized.

c) Integration: Third party modules such as payment gateway, delivery service module are integrated to provide users with more options and benefits. The product content and catalog are integrated which either the seller / store back end team can upload.

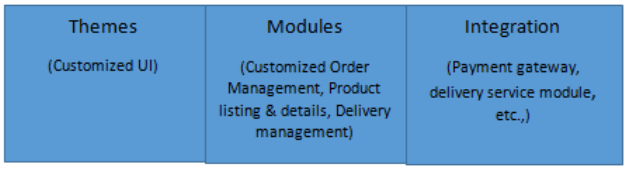

Fig 6: Customized Standard e-commerce platform

III. INTEGRATION ACTIVITY

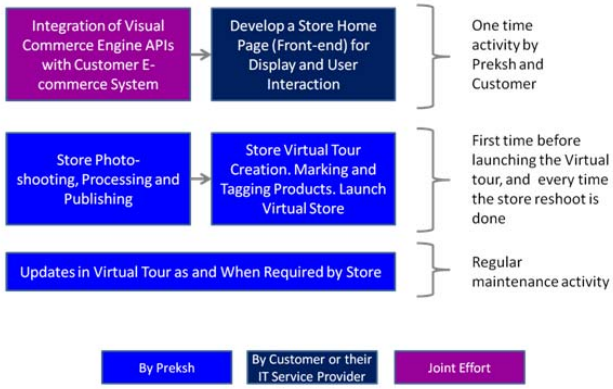

Fig 7: Activity block diagram

There are certain steps taken convert an ecommerce system to visual ecommerce system. If a customer requires a visual ecommerce engine to be integrated, they have to follow some simple steps mentioned above in the fig 7. Around 6 APIs are involved to complete the integration of Visual Commerce Engine. This is defined in the fig 8 as shown below.

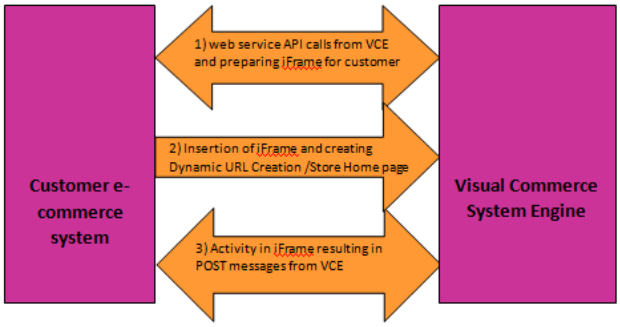

Fig 8: API integration

IV. SUMMARY

Using our engine/system, stores can rebuild their trust and relationship with customers offline as well as online. This system uniquely combines the features toolset and adds functions so to provide a competitive Visual-commerce Shopping System. It realistically simulates the physical store's intended atmosphere, environment, and customer's shopping experience, which uniquely brands that store. Stores can update the products and their details any time using our system. This provides an interface where online stores can interact with customers to give them a better service and keep a long term engagement with customers.

Through our AR Solution, the artisans & craftsmen and unique processes involved can be effectively showcased (using relevant audio-video, images or text information), along with their products, for effective product marketing. Institutional and Government support to such an initiative will further help to preserve the culture and promote artisans & craftsmen by enabling them to get better rewards for their tireless efforts.

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